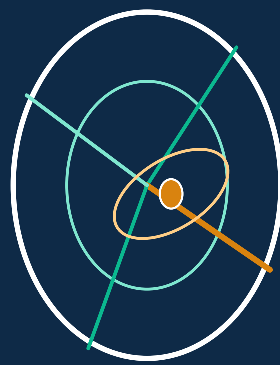
ARES ATAK

RF propagation · passive geolocation · distributed sensing · 3-D globe · ATAK

From sensing to situational awareness — one platform.

Locate an emitter across a mesh of sensors → predict its coverage on real 3-D terrain → push it to every operator's ATAK, and chat about it — air-gapped if you have to. The terrain physics is the actual ITS Longley-Rice; the geolocation is a maximum-likelihood fix with a geometry-correct error ellipse; the radios range from one dongle to a coherent DF array to a whole networked team of Ares nodes on a MANET.



Terrain RF propagation
ITS Longley-Rice (ITM) + ~12 models, raster

Passive geolocation / DF
ML fix + error ellipse · TDOA/FDOA · array MUSIC

Single-channel DF
RSS · Doppler-CPA · synthetic aperture · phase int.

Auto PTT identification
DMR/P25/TETRA/NXDN/D-STAR/YSF/M17 + auto-decoder

UAS / FPV video decode
NTSC/PAL FM video · colormaps · snapshot/record

3-D globe
CesiumJS — RF on real terrain & buildings

ATAK / TAK
CoT out: LoBs · fixes · GeoChat — UDP/mcast/TCP/TLS

Distributed sensing
MANET peer fusion, mesh-signed · group chat

Self-hostable, offline
Kali-ready installer · Jetson · laptop · Pi · cloud

Open source · runs fully offline · feature-equal to the SOOTHSAYER ATAK plugin against an Ares server — plus DF.

Get started

Install — ./install.sh (Linux/macOS) · install.bat (Windows) · air-gapped: ./install.sh --offline-bundle <dir>

Run — ./start-backend.sh (server :8000) · ./start-web.sh (browser UI :3000) · ./start-desktop.sh (Electron) · docker compose up -d

Explore — http://localhost:8000/docs (interactive API) · docs/Ares.md · docs/DEPLOYMENT.md · cd backend && python -m tests.test_authoritative

Then read the companion 'Ares Tutorial' PDF (a slide-by-slide walkthrough). New to programming? Ask the project chat for the learning roadmap.

What Ares does

Six things most products keep separate — in one self-hostable stack.

Terrain RF propagation

The authoritative ITS Longley-Rice (ITM) — the SPLAT! / Radio-Mobile / FCC algorithm — plus ~a dozen empirical & ITU models, real diffraction (Deygout / Bullington / Epstein-Peterson / Giovanelli), atmospheric and space-weather corrections, a radar equation, 20+ analytic antenna patterns and measured-pattern import (NSMA/Planet MSI, NEC-2). Coverage as a radial heatmap or a per-pixel raster; point-to-point link budgets with terrain profile, Fresnel zone and LOS-obstruction; multisite / best-server / best-site; interference & EMCON; ray-trace; route; MANET coverage.

Passive geolocation / DF

Maximum-likelihood bearing-only triangulation with a covariance-derived (geometry-correct) error ellipse, GDOP and an EKF emitter track; TDOA/FDOA hyperbolic multilateration; phase-interferometry / MUSIC / Capon array DoA with the CRLB. Bearings are terrain-capped via the propagation engine — DF that respects mountains. Three compass modes (Absolute LOB · Relative LOB · clock position) plus a guided calibration procedure.

SDR console & spectrum

Single-channel SDRs monitor a spectrum / decode audio; multi-channel SDRs (declare the channel count) also produce lines of bearing. A DF panel gives stacked spectrum viewers (scroll-zoom, fixed y-axis) with a ▦ waterfall, a live LoB compass, and the DF options (tuner, active-power threshold, gain/AGC, demodulate-and-listen). SoapySDR drives real RF when installed; an audio-decode bridge dispatches DMR / P25 / TETRA / NXDN / … to op25 / dsd-fme / sdrtrunk when present.

3-D globe & offline data

A CesiumJS 3-D globe alongside the Leaflet 2-D map: coverage, LOS, Fresnel zones, antenna lobes and obstruction markers on real heightmap terrain; KMZ import/export persisting across 2-D ↔ 3-D; offline data packs (terrain / OSM / imagery / buildings / clutter) with a provider chain that grows the pack online; a 'you-are-here' GPS marker.

ATAK / TAK integration

Cursor-on-Target out (UDP / multicast / TCP / mutual-TLS) — LoBs as drawn routes, fixes as intel ground points with a CEP circle, chat as GeoChat — and a CoT receive listener so GeoChat from ATAK joins the same conversation. Offline data-pack and radio-template management. An open ATAK-CIV plugin (SDK-blocked for build/signing) for SOOTHSAYER-style parity plus DF.

Distributed sensing & chat

Multiple SDRs on one box cross-fuse automatically; over a MANET, peer Ares nodes share LoBs / fixes / chat — mesh-signed (HMAC over a shared secret) and loop-safe (dedup, hop-count) — so the union of every node's bearings is fused on every node. Group chat with rooms, bridged to ATAK GeoChat. HF circuit prediction (ITU-R-P.533-style: hops, MUF/FOT/LUF, absorption, reliability) and real-SGP4 satellite visibility round it out.

THE LIVE LOOP

How it works

From a radio (or several, networked) to a fix, a coverage prediction, and ATAK.

▸ 1 · Sense

An SDR (KrakenSDR / Matchstiq X40 / generic — or a manual operator) produces a line of bearing; a coherent array's snapshot is turned into one by interferometry / MUSIC.

▸ 2 · Fuse → Fix

All bearings at the same frequency, from this node and every mesh peer, feed the ML solver: a maximum-likelihood emitter position with a geometry-correct error ellipse, GDOP, and an EKF track.

▸ 3 · Predict

Optionally a coverage run reruns from the computed emitter location (ITS Longley-Rice on real terrain) — so you see its predicted reach, updating live as it's located.

▸ 4 · Push

LoBs, the fix (with its CEP circle) and chat go out as Cursor-on-Target — UDP / multicast / TCP / mutual-TLS — appearing natively in ATAK / WinTAK / on a TAK Server.

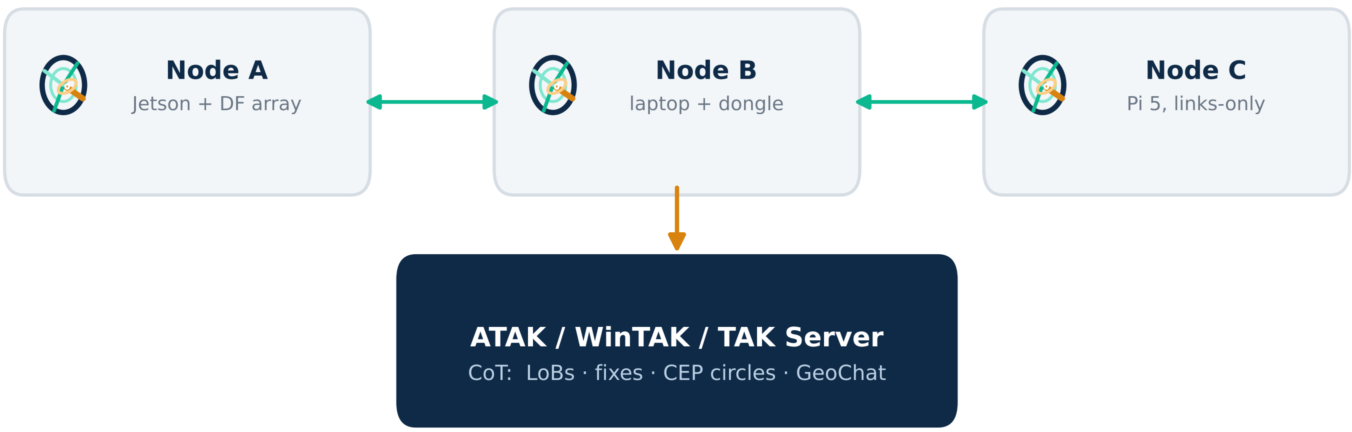
▸ 5 · Share

Over a MANET every Ares node re-broadcasts what it learns (signed, deduplicated, hop-bounded) — so the fused picture and the group chat live on every node, not just the one with the antenna.

▸ 6 · Map

It renders on the 2-D Leaflet map and the 3-D CesiumJS globe: coverage heatmaps, LoB fans, error ellipses, suspected-emitter markers, KMZ overlays, your GPS position — online or offline.

One server — or a mesh of them



Each node shares LoBs / fixes / chat over the mesh — HMAC-signed with a shared secret, deduplicated and hop-count-bounded; the fusion runs on every node, so losing one node loses a sensor, not the picture. Same physics throughout; offline-first, with online auto-fetch when reachable.

Where Ares stands

No single product spans this — Ares ties the layers together, openly.

Capability	Ares	Closest existing tools
Terrain propagation	**ITS Longley-Rice + ~12 models, raster**	CloudRF · SPLAT! / Radio Mobile (ITM only)
Geolocation / DF	**ML fix + error ellipse, TDOA/FDOA, MUSIC**	KrakenSDR doa app (DoA + basic triangulation)
SDR spectrum / audio	**spectrum + waterfall, decode bridge**	SDR# / SDRtrunk / op25 (no propagation, no fix)
3-D globe	**CesiumJS, RF on real terrain**	STK (orbits, not RF coverage)
Offline / air-gapped	**data packs + provider chain; Jetson/Pi**	SPLAT! (yes, 1990s UI); CloudRF (SaaS-bound)
ATAK / CoT out	**LoBs/fixes/GeoChat — UDP/mcast/TCP/TLS**	SOOTHSAYER (CoT via a commercial plugin)
Distributed (MANET)	**peer fusion, mesh-signed, + group chat**	— (nothing comparable, open or closed)
Licensing	**open source, self-hostable**	CloudRF/STK commercial · SPLAT!/doa-app open

Deployment targets

- **NVIDIA Jetson Orin —**
  the full server with a GPU (CuPy multisite / Monte-Carlo); the 'Ares-in-a-box' vehicle node.
- **Rugged x86 laptop —**
  the full server, CPU or eGPU; pair it with a 500 GB+ SSD for the data packs.
- **Raspberry Pi 5 —**
  a 'links-only' node: P2P link mode, RF-link / Co-Opt, DF, HF, space weather.
- **Cloud VM —**
  a shared server; leave ARES_AUTH on (the default whenever it isn't bound to loopback).

Ares — and honest about the rest

Reference-grade: the ITS Longley-Rice ITM port (lrprop / avar / adiff / ascat / alos, 7 climate zones, NTIA Report 82-100); the ML / Stansfield triangulation + covariance error ellipse; TDOA/FDOA (Chan); multi-baseline phase interferometry + CRLB; MUSIC / Capon / Bartlett; SGP4; the ITU-R-style HF circuit; CoT / GeoChat I/O including mutual-TLS; the HMAC-signed MANET fusion.

Still approximate / pending hardware: the HF foF2 is a parameterised model, not the CCIR/URSI coefficient maps; ITM isn't yet bit-validated against NTIA's C reference; hardware-in-the-loop (KrakenSDR → fix → CoT) and a live multi-node mesh are code-exercised but untested on real RF; the ATAK plugin is SDK-blocked for build/signing. All of it is spelled out in docs/Ares.md.